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Coupled simulations of water flow from a field-investigated glacial till slope using a quasi-two-dimensional water and heat model with bypass flow

Bengt Espeby

Department of Land and Water Resources, KTH, S-10044 Stockholm, Sweden

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ABSTRACT

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Substantial field investigations of soil physical properties and stratification in a forested slope (10° slope) covered with glacial till revealed that macropores in the form of old root channels as well as coarse structures in the form of well-sorted layers dominate a very permeable upper solid horizon. Rapid response and quick recessions during snowmelt and heavy rains in 1986 were observed in the runoff from the slope. Based on field tracer experiments it was found that such macropores and macrostructures played an important role on these occasions. In order to verify these findings one-dimensional water and heat models were coupled in a serial manner to simulate the formation of runoff from the slope, using a quasi-two-dimensional approach. Both a strict Darcian concept and a Darcian concept with a simple bypass flow concept introduced were tested. The drainage gadient in the model was made equal with the angle of the slope. Coupled slope simulations, with water retention properties and hydraulic conductivities taken from three different levels on the slope, indicated that the conditions in the lower region of the slope were most important in explaining the discharge rate. With a shallow groundwater table in the lower region of the slope and low hydraulic conductivity of the deeper layers, rapid water flows are routed to the uppermost layers where the conductivity is higher. Most of the flow is well described by Richards' equation, although smaller peaks cannot be represented for small rain events, when the measured runoff and recession showed a more rapid response than that simulated, however, the introduction of a simple bypass flow improved the ability of the model to simulate the observations. Much of the simulated surface runoff generated in the Darcian simulation during the winter of 1986, could be diverted through a frost layer in the humus horizon and in the humus-impregnated mineral horizon with a silty-sand character, down to a more conductive and porous layer. Eighty percent of the water flow in the bypass flow simulations was conducted in the upper 40 cm of the soil with peak intensities in the more porous upper 20 cm where the macroporosity was large.

INTRODUCTION

Water movement in forest soils and forest hill slopes has long been the subject of research. The objectives have been many, e.g. to clarify runoff

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generation (Kirkby, 1978), groundwater recharge (Troedsson, 1955; Johansson, 1986), the influence of vegetation on flow patterns (Greminger, 1984), leakage of nutrients from forest soils (Troedsson, 1955), and erosion processes (Leopold et al., 1966 cited in Kirkby, 1978).

Many observations of the quick response of groundwater levels and soil moisture to rainfall or snowmelt events have been explained by rapid drainage of free water via macropores present in the soil, e.g. increasing soil moisture in deeper horizons after heavy rains without an increase in soil moisture in overlying layers (Andersson, 1988), quick response in rising groundwater tables in Norwegian tills (L.A. Kirkhusmo, personal communication, 1989), and delayed simulated drain discharge compared with measured drain discharge (Jarvis and Jansson, 1989 — not a forest soil). In a study of stream flow generation in a steep (average 34°) forested headwater catchment on South Island, New Zealand, Mosley (1979) concluded that macropores had a dominant influence on the mechanism of storm runoff generation. In a very thorough quantitative investigation of flow patterns in a 40° (275 m^2) sloping experimental plot using 500 tensiometers and 50 neutron access tubes, Greminger (1984) concluded that an almost simultaneous response in the vertical direction was caused by an interconnected macropore system. He also concluded that the lateral component of flow was significant in two horizons even without complete saturation. Other, more common, observations of quick response in measured groundwater tables have been made in heavy clay soils where stable cracks have been formed after a long period of drought (Jarvis, 1989). In a comprehensive review article concerning macropores Beven and Germann (1982) also give thorough coverage of the definitions, experimental evidence and theoretical aspects of water-conducting macropores in soils.

Field studies of a glacial till slope in Sweden revealed rapid response in the runoff and quick recessions during periods of snowmelt and heavy rains. Discharging macropores and macrostructures were an important factor during these events (Espeby, 1988, 1990a). This confirms thorough in situ measurements of the hydraulic conductivity in Norwegian till (Haldorsen et al., 1983) which showed that more than 90% of the gravitational water movement occurs along fractures, fissures and channels, i.e. macropores. In addition to macropores, frost conditions in soils during the snowmelt also control water flow. Snow often starts to melt before the soil is thawed. Many researchers report that infiltration starts while the soil is still frozen and is more or less restricted to macropores (e.g. Komarov and Makarova, 1973; Steenhuis et al., 1977; Lundin, 1989). Roberge and Plamondon (1987) report, in a study of snowmelt runoff pathways in a forest slope, that water transported rapidly by throughflow in macropores or pipes was more acidic than the ground water. The snowmelt runoff was later modelled using a variable source area simulator (VSAS2) by Prévost et al. (1990). Kane and Stein (1983) point out that the infiltration rates during snowmelt in a seasonally frozen soil are very much dependent on the initial moisture content of the soil prior to ablation.

However, apart from these observations there have been few hydrological investigations made on slopes covered with glacial till. The prevailing conditions with unsorted texture, large boulders and discontinuous lenses with sorted texture have made investigations difficult to perform and the results hard to extrapolate to other catchments with similar conditions. Most of the studies performed on till have been of a more descriptive nature. Few theoretical studies on till catchments using transport models have been reported. However, among several methods used to determine groundwater recharge in sandy till aquifers in southeastern Sweden, Johansson (1986, 1987) found that a one-dimensional physical water and heat flow model gave the best representation of outflow measurements in his study catchments. It is here important to mention some theoretical studies considering physically based modelling of hillslope flow. Fipps and Skaggs (1989) investigated the influence of slope on subsurface drainage of hillsides using a two-dimensional form of Richards' equation for combined unsaturated and saturated flow. They found that small slopes ($< 0.15\,\mathrm{m\,m^{-1}}$) had little effect on drainage flow rates and water table depths in the centre of the region. Binley et al. (1989a, b) used a fully three-dimensional model to explore the effects of a random pattern of saturated hydraulic conductivity on runoff production from a $100\,\mathrm{m} \times 150\,\mathrm{m}$ hillslope. They concluded that peak discharge and runoff volumes were generally increased by the presence of increasing heterogeneity. For the case of low-permeability soils, characterized by surface flow domination of the runoff hydrograph, single effective parameters were not found to be capable of reproducing both subsurface and surface flow response unlike high permeability soils where effective parameters were found to reproduce reasonably the hillslope hydrograph.

Despite many indications of macropore flow in glacial till, few studies (if any) have considered macropores when modelling water movement. The objective of this paper is to demonstrate how different horizons and zones of a slope covered with glacial till respond to natural meteorological conditions during a 12 month period, by using a one-dimensional physically based water and heat transport model, SOIL (Jansson and Halldin, 1979). This paper describes how SOIL was used in a quasi-two-dimensional way by incorporating additional boundary conditions to simulate horizontal water flow at different soil horizons. The work attempts to answer questions such as: Can a Darcian approach with a single pore velocity simulate water movement in this kind of soil or is it necessary to introduce a bypass flow in the model in order to simulate the flow dynamics? What part of the slope is regulating the runoff dynamics? Is frost influencing the water flow?

SITE DESCRIPTION

The selected catchment Lund (2.3 ha) is located 35 km northwest of Stockholm (59°29′ N and 17°48′ E) in the south-central part of Sweden (Fig. 1). It is situated below the highest found shoreline (formed after the latest ice age when the area was covered by meltwater) on a northwest slope (10° slope) covered by glacial till with an altitude ranging between 20 and 45 m a.s.l. Eighty-five to ninety percent of its surface is covered with 60–70-years old coniferous forest with spruce (*Picea abies*) and pine (*Pinus sylvestris*) the dominant tree species. The upper part of the study area consists of outcrops of rock covered with moss and lichen (Espeby, 1989). The rock area is delimited by a boulder region which is approximately located at the border of the forest. Using the general term of Gregory and Walling (1973) the site is a first-order stream catchment.

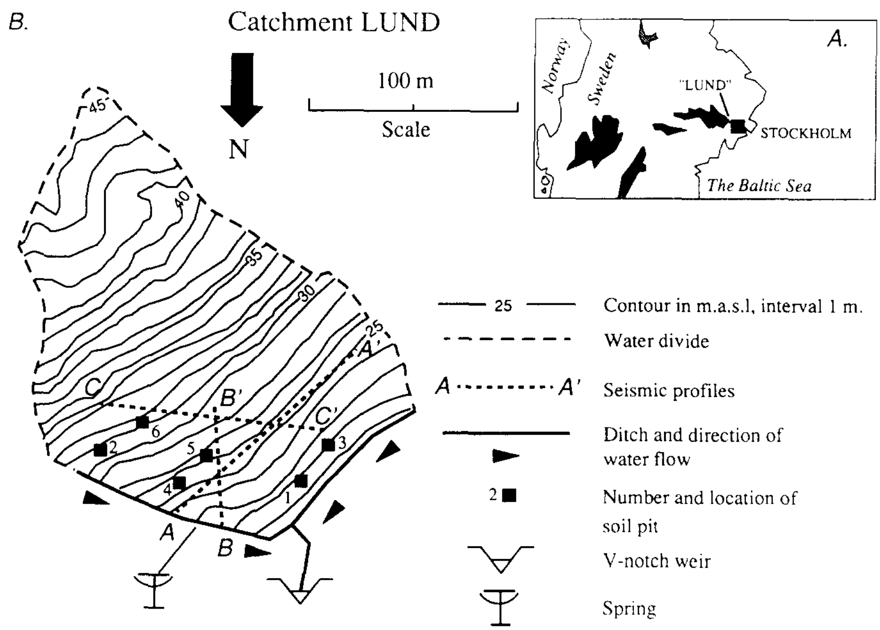


Fig. 1. Catchment Lund. (A) Location in south-central Sweden. (B) Detailed map of the sloping catchment with pit locations and seismic profiles.

MATERIAL AND METHODS

The upper 50 cm of the soil materials were carefully investigated at six different locations on the slope (Fig. 1(B)). At these six locations, water retention characteristics, saturated hydraulic conductivity, texture and bulk porosity were measured in the laboratory.

Seismic soundings along three profiles were used to determine the overburden thickness in the area (Espeby, 1989). The upper 0.5–1 m of the overburden consists of a loose till covering a compact basal lodgement till with a thickness of 0.5–4 m. The average overburden thickness is 3–4 m with a maximum thickness of 6 m observed in the bottom region of the catchment. The bedrock surface undulates and a ridge in the bedrock close to the ground surface is found in the lower region of the slope. Higher seismic velocities in the lower region of the slope indicate a more compact till (see Fig. 2).

Soil properties

The schematic principle soil stratification of five major soil horizons (based on the detailed investigations from the six soil pits) is presented in Fig. 3. The organic strata of litter and humus is covering a wave-washed beach gravel and sand layer (formed as a result of uplift of the land after deglaciation) which has a high saturated hydraulic conductivity, $K_s(10^{-4}\ \mathrm{m\,s^{-1}})$. Underlying this horizon is a fine textured horizon with a silty-sand character (average K_s $10^{-6}\ \mathrm{m\,s^{-1}}$). A clayey silty-sandy till underlies these layers, having a very low saturated hydraulic conductivity, average $2.5 \times 10^{-8}\ \mathrm{m\,s^{-1}}$ (Espeby, 1990b).

Total porosities of the five upper pedologically changed soil layers

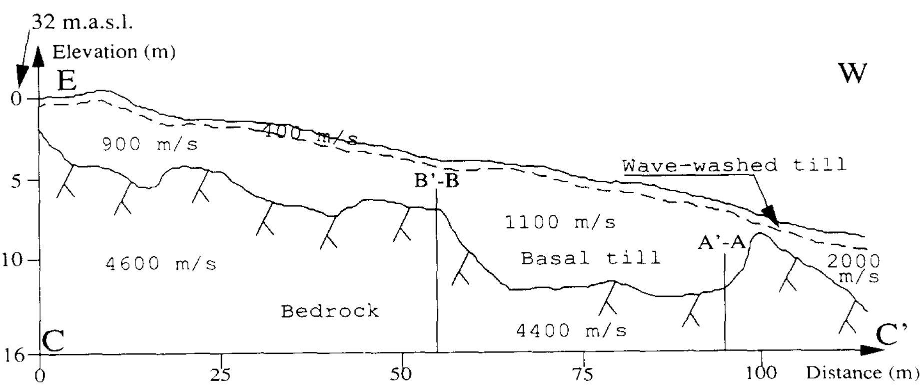


Fig. 2. Cross-section C–C′ along the soil-covered part of the slope in Lund. Approximate seismic velocities (in $\mathrm{m\,s^{-1}}$) are given for the wave-washed till, basal till and the underlying bedrock and the locations of two other seismic profiles are noted.

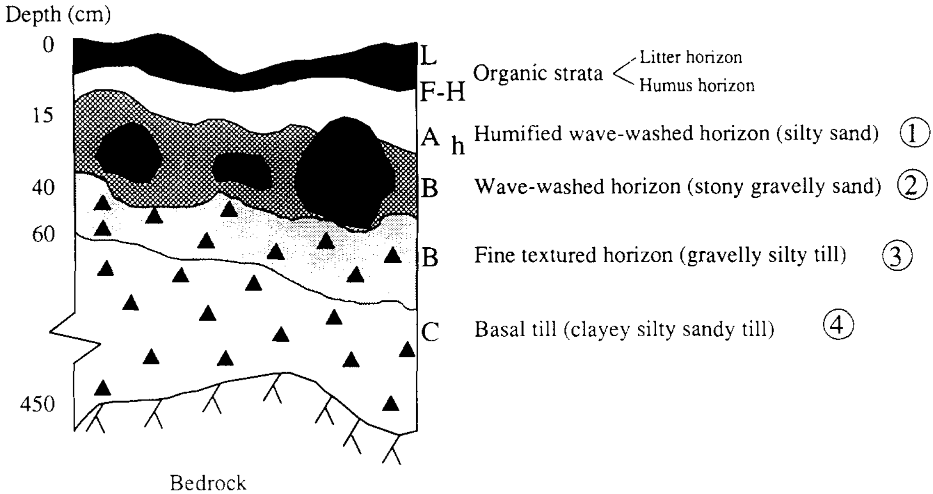


Fig. 3. Schematic principle stratification. The horizon numbers refers to Table 1 and Fig. 4.

(including the organic strata of litter and humus) have been determined with a new photogrammetric method at two of the above mentioned pits (Espeby, 1986) (see Fig. 4). The organic strata (litter and humus) have a very high total porosity (93–95 vol.%). The humified mineral horizon increases in thickness and the number of macropores in the form of old root channels increases towards the lower part of the slope. This is also reflected in the increasing porosity towards the lower part of the slope. Approximately 85% of the roots are located in the upper 25–30 cm of the soil. The boulder content is high throughout the whole slope with many of the boulders reaching the soil surface and thereby influencing the infiltration pattern. Saturated hydraulic conductivities, K_s, based on constant head permeameter tests on undisturbed vertical soil cores with a diameter of 70 mm (Espeby, 1989, 1990b) are presented in Fig. 4. The trend of decreasing conductivity with depth is a common feature in this kind of soil (Lundin, 1982; Johansson, 1986, 1987; Espeby, 1989, 1990b) but the spatial variability also increases with depth (Table 1). This variability in K_s (up to two orders of magnitude) is present horizontally as well as vertically (when measured on cores taken from individual pits). Some of this variability can be explained by the presence of the above-mentioned macropores found throughout the slope.

Water retention properties were determined in the laboratory using soil cores (70 mm diameter, 50 mm and 100 mm height) collected vertically in the upper 1 m of the overburden at pits Nos. 1, 4 and 6 (Fig. 5). The laboratory-measured water retention has been fitted to an analytical expression of the water content/tension relation proposed by Brooks and Corey (1964) in the

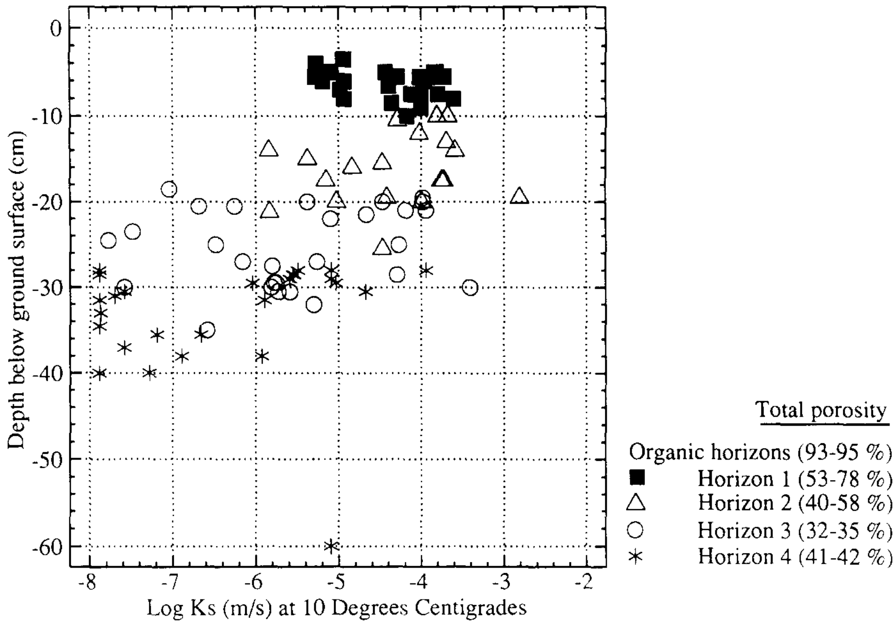


Fig. 4. Saturated hydraulic conductivity, log K_s, from four mineral horizons vs. depth for 100 samples from six pits in Lund, and total porosity of each horizon (including the organic horizons).

suction range between 0.10–150 m of water. The saturated hydraulic conductivity (matrix conductivity) from these three pits was estimated from the measured total saturated hydraulic conductivity (including the effect of macropores) based on a decreasing amount of macropores with depth and the present mineral texture of each horizon (Espeby, 1989), i.e. less difference between total saturated conductivity and matrix conductivity with depth. Unsaturated hydraulic conductivities were determined using the analytical expression of Mualem (1976), modified by Jansson and Thoms-Järpe (1986)

TABLE 1

Descriptive statistics of saturated hydraulic conductivity, log K_s(m s^{-1}), related to the mineral horizons (ref. Fig. 3) from 100 core samples taken at six different sites in the Lund area

Horizon	Nos. of samples	Mean	Standard deviation
1	27	−4.42	0.56
2	19	−4.36	0.82
3	28	−5.52	1.20
4	26	−6.43	1.25

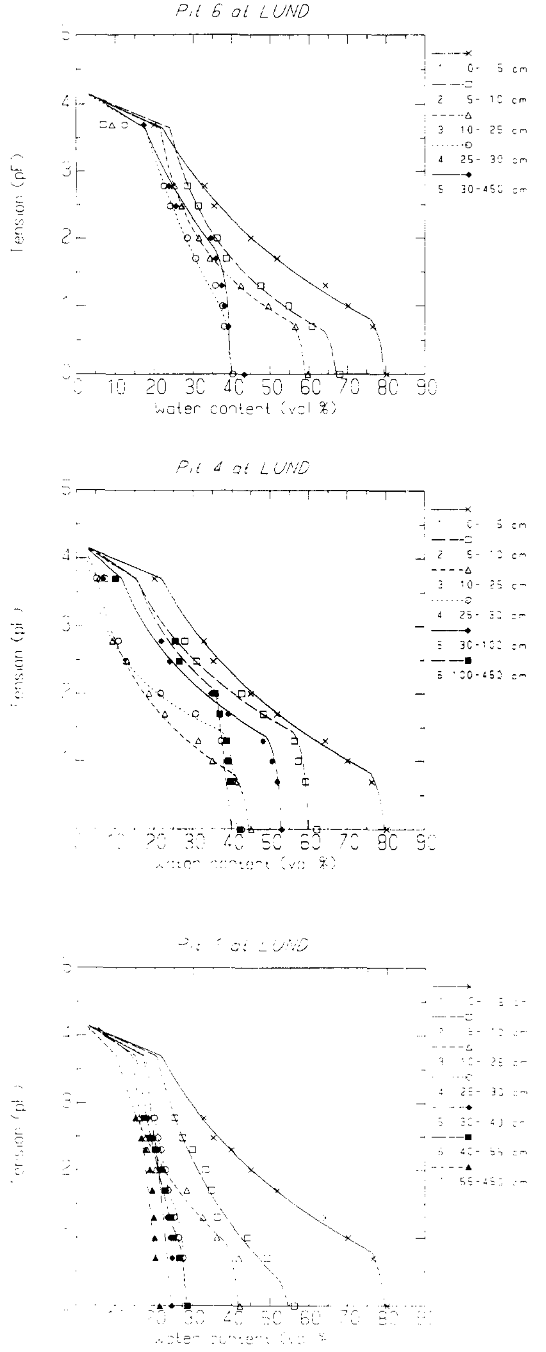


Fig. 5. Water retention properties from pits Nos. 6, 4 and 1. The water retention properties from the lowest layer (unaffected basal till) of all three pits have been extrapolated down to the bedrock surface, to the depth of 4.5 m.

to account for the contribution by macropores, and based on the expression of effective saturation defined by Brooks and Corey (1964).

The water retention properties from pits with soil layers that had substantial amounts of large stones and/or boulders were adjusted to account for the dead porosity, i.e. the pF-curve from the specific soil layer was altered in the low pressure range (> -10 kPa) to account for smaller porosity but with the same drainage characteristic. This was done by decreasing the total porosity at saturation, the pore-size distribution index and the air entry value (Brooks and Corey, 1964) but keeping the same residual water content. Volume measurements with a photogrammetric technique (Espeby, 1986, 1989) indicated a reduction in the total porosity of 5–15% for stone-rich layers. The wave-washed horizon and the basal till in the upper part of the slope (pits Nos. 2 and 6) have especially large numbers of stones and boulders.

Measurements

Runoff from the slope was recorded from spring 1986 until the beginning of June 1987 by means of a 60° V-notch weir placed in a ditch draining the catchment (Fig. 1). It was not possible to install groundwater tubes with available methods.

Air temperature was measured on the slope at 0.5 m above ground (soil pit No. 1, see Fig. 1). At two places on the slope, soil temperatures (soil pits Nos. 1 and 4) were continuously measured at 7, 20 and 36 cm depth from autumn 1985 to spring 1987. Snow surveys were performed in the catchment during the snowmelt periods in 1985–1987.

Precipitation data were taken from the Swedish Meteorological and Hydrological Institute (SMHI) Synoptical station 9732 at Sätra Gård (59°32′ N and 17°51′ E), 3 km north of Lund and situated at the same altitude as the monitored catchment. Daily means of global radiation, wind speed and cloudiness were taken from Bromma Airport (59°21′ N and 17°57′ E) and air temperatures from Ultuna (59°49′ N and 17°39′ E) in order to get temperatures representative of forested conditions. The average yearly precipitation is 610 mm and the average yearly runoff is 200–210 mm in this part of Sweden (Engqvist and Fogdestam, 1984).

The SOIL model

The model used in this study was a physically based one-dimensional water and heat model called SOIL. The model was originally developed by Jansson and Halldin (1979) for the Swedish Coniferous Forest Project to be utilized for simulating water and heat flow in forest soils. Since then the model has been utilized on various kinds of soils from heavy clay soils and other agricultural soils (e.g. Jansson, 1987b; Jansson and Gustafsson, 1987; Lundin, 1989; Thunholm et al., 1989) to more unsorted glacial tills (e.g. Johansson, 1986, 1987). For a detailed technical description of the model see Jansson and Halldin (1980). Recent development of the model as well as its availability is found in Jansson (1987a).

The main part of the model describes water and heat transfer in a vertical soil profile using two partial differential equations solved by an explicit forward finite difference scheme (the Euler method). The two flow equations, for water and energy, describe the change in water content and temperature over time and depth. The unsaturated water flow is given by Darcy's law in combination with the continuity equation (Richards, 1931):

$$\frac{\partial \theta}{\partial t} = -\frac{\partial}{\partial z}\left[k_{\mathrm{w}}(\theta)\left(\frac{\partial \psi_{\mathrm{w}}}{\partial z} + 1\right)\right] - s_{\mathrm{w}} \tag{1}$$

where the hydraulic conductivity, k_{w} varies with the water content, θ, and ψ_{w} is the soilwater tension, and s_{w} is a sink-source term accounting for root water uptake. The water retention characteristic is treated with the analytical form derived by Brooks and Corey (1964) not accounting for any hysteresis effects.

Heat flow is given by Fourier's law (Carslaw and Jaeger, 1959) in combination with the continuity equation:

$$\frac{\partial (CT)}{\partial t} - L_{\mathrm{f}}\rho_{\mathrm{i}}\frac{\partial \theta_{\mathrm{i}}}{\partial t} = \frac{\partial}{\partial z}\left(k_{\mathrm{h}}\frac{\partial T}{\partial z}\right) - C_{\mathrm{w}}\frac{\partial (T_{\mathrm{w}}q)}{\partial z} - s_{\mathrm{h}} \tag{2}$$

where C and C_{w} are the heat capacities of the soil and water, respectively, q is the flow of water with a temperature of T_{w}, and k_{h} is the heat conductivity. The left-hand term describes the change in sensible heat, CT, and the latent heat of freezing, L_{f}, to ice, θ_{i}. The first term on the right-hand side describes the divergence of the heat flow, the second term the convective heat flow and third term, s_{h}, describes a sink-source term accounting for heat extraction.

A maximum of 22 compartments of soil layers can be used in the model (Fig. 6). Compartment thicknesses are chosen to correspond to the morphological structure of the soil and to meet the numerical requirements of the Euler method. Thus the smallest compartments are needed near the soil surface to account for the more pronounced variations in soil properties as well as state variables found here. Soil properties and state variables are calculated linearly between midpoints of consecutive model compartments.

Appropriate upper boundary conditions of the model are given by submodels for precipitation, snow and frost dynamics, interception storage and evapotranspiration. The daily amount of snowmelt is calculated from a heat sum, a factor accounting for the influence of solar radiation and the

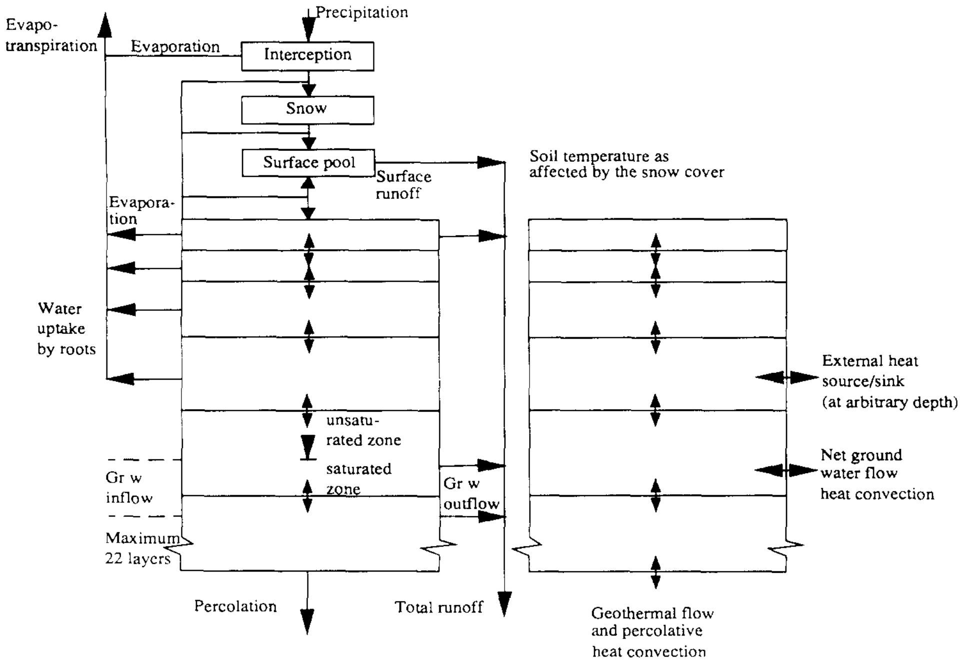


Fig. 6. Structure of the SOIL model (from Jansson and Gustafsson, 1987).

surface heat flow (Jansson and Halldin, 1980). The lower boundary condition in the water flow equation can be considered as horizontal groundwater outflow, which may be affected by drainage level; conceptually this can be a drainage pipe in an agricultural field or a groundwater table sloping towards a discharging stream in a forest. The lower boundary can also be represented by free drainage to an infinite groundwater body. The location of the groundwater table is calculated as the level where soil moisture tension is nil (cf. Johansson, 1986). Initial values of tensions, and the location of the groundwater table are used to calculate initial storages of water. An initial equilibrium tension profile above the groundwater table is assumed (i.e. no vertical flow). From each of the model compartments, free water exceeding a given retention threshold is lost to infiltration, either vertically to an infinite groundwater table or horizontally if the groundwater table is located in the compartment immediately below the compartment with excess water. A geothermal heat flow or a variable temperature is used as the lower energy boundary.

Bypass flow in the SOIL model

A rigorous model of macropore flow would use a two-domain concept (e.g. Van Genuchten and Wierenga, 1976; Jarvis, 1989), solving the storage and

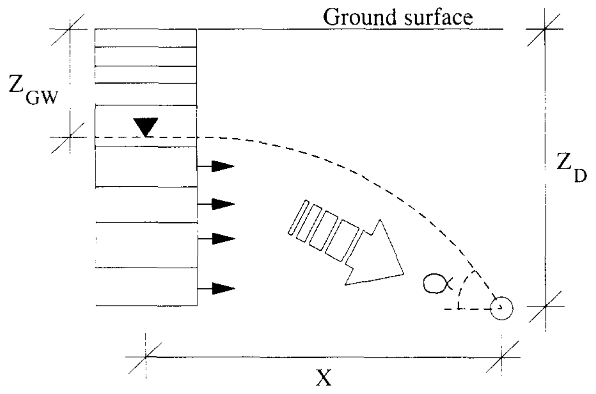


Fig. 7. The conceptual representation of the drainage gradient in the model.

flows in the matrix and the macropores with two coupled differential equations. However, a simpler empirical way of handling macropore flow is used in this study. The SOIL model uses a bypass flow concept where the sorption capacity of the matrix (S_m) is scaled (Jarvis and Jansson, 1989). At each timestep, water entering any soil compartment at a rate greater than S_m is routed as bypass flow to the next compartment vertically downwards in the soil profile. A simplified form of the mass balance equation in one compartment using Darcy's law is thus:

$$q_2 = \max(q_1 - S_m, q_d) \tag{3}$$

were q_1 is the water inflow to the compartment, q_2 the water outflow from the compartment, and q_d is the flow calculated by Darcy's equation. The sorption capacity of the matrix is given by:

$$S_m = K_s \log_{10}\psi \, \Delta z \, A_{SC} \tag{4}$$

where Δz is the thickness of the compartment and A_{SC} (< 1) is a lumped non-dimensional parameter, which partly reflects the size and geometrical configuration of aggregates and the degree of saturation in macropores (Jarvis and Jansson, 1989). Although the scaling factor, A_{SC}, is dependent on the soil physical properties, its value has to be determined by a fitting technique. In this model, A_{SC} is effective for the whole soil profile and cannot be set for single layers, i.e. each model compartment has the same A_{SC}-value.

Slope simulation

To represent a draining slope in a one-dimensional vertical model it is necessary to introduce a driving force in the horizontal direction by introducing a drainage level in the model. The slope gradient (α) is then calculated as the quotient of the difference between the groundwater level (Z_{GW}) and the

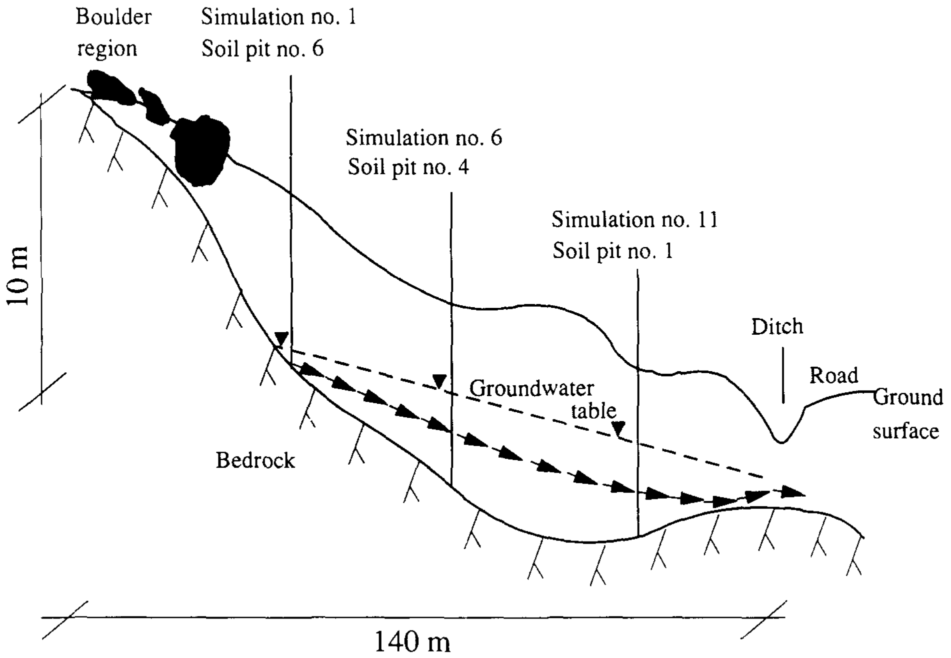


Fig. 8. The conceptual representation of one realization of coupled slope simulations with initial groundwater tables at simulations Nos. 1, 6 and 11. Each arrow represents the saturated outflow from the model segments in each simulation (cf. Fig. 7).

drainage depth (Z_D) and a horizontal distance (X) (cf. Fig. 7):

$$\alpha = \frac{Z_{GW} - Z_D}{X} \tag{5}$$

The values of drainage depth (Z_D) and horizontal distance (X) were chosen so that the drainage gradient would be close to the average slope of the catchment when the groundwater level is at the ground surface, i.e. 5 m for Z_D and 70 m for X.

Coupled slope simulation

To simulate different flow regimes within the slope, additional boundary conditions were added to the model to allow horizontal input/output of water at different horizons and segments (cf. Fig. 6), i.e. simulating a two-dimensional problem using coupled simulations from a one-dimensional model. The first model realizations were pure Darcian flow simulations without bypass flow. Each realization consisted of 15 coupled simulations with soil properties taken from three different soil pits (pits No. 6 in the upper part of the soil covered part of the catchment, No. 4 in the middle part of the catchment, and No. 1 from the lower part of the catchment) (Figs. 1, 5 and

8). Soil properties from pit No. 6 were valid for simulations Nos. 1–5, properties from pit No. 4 for simulations Nos. 6–10, and soil properties from pit No. 1 were valid for simulations Nos. 11–15.

Parameter choice and simulation technique

The model was compared with measurements from 1.5 years runoff data between 1 September 1985 and 31 December 1986. Simulated runoff (i.e. accumulated drainage water from the model profile) was compared with the measured runoff (in mm), calculated for the whole catchment area (2.3 ha). Evaporation parameters were taken as averaged values from two field evaluations using the SOIL model in one spruce forest in Svartberget, northern Sweden and in one beech forest in Rävsmåla, southern Sweden (L.-G. Nordén, personal communication, 1989).

The saturated hydraulic conductivities of the surface horizons were point values from single cores, whereas the saturated hydraulic conductivities of the deepest layer at each level of the slope (in each slope realization) were average values from a statistical analysis (Espeby, 1990b). The average conductivities of the deepest layer at the three pit locations were 0.036 cm h^{-1} for the top level, 0.114 cm h^{-1} at the middle level, and 0.009 cm h^{-1} at the bottom level (cf. Fig. 4 and Table 1).

A linear equilibrium profile was used for the initial groundwater tables in the slope. This was in turn based on the final groundwater tables from several initial slope realizations where the first realization used an initial groundwater table of -4.0 m at all levels of the slope. After each slope realization, the final values of the groundwater tables in each single simulation were used as initial values of the groundwater level in the following slope realization. The drainage distance (X) and the drainage depth (Z_D) were kept constant in all 15 simulations at 70 m and -5 m, respectively, resulting in a gradient of 7% with the groundwater table at the soil surface.

Measured soil temperatures before, during and after the snowmelt, were compared with simulated temperatures in order to get a reasonable agreement of the heat flow during this period. The snow survey data from the closed canopy forest were used to calibrate the snow dynamics of the snow submodel in SOIL.

RESULTS AND DISCUSSION

General hydrology 1986

The snow surveys performed in open areas and in closed canopy forest during the monitoring period 1985–1987 showed a clear difference with a 50% higher snow storage in the open areas than in the dense forest (Espeby, 1990a). The usual explanation for this effect is that it is caused by increased deposition at open-forest boundaries (Berris, 1984, cited in Harr, 1986). Recent research results indicate the possibility of higher interception losses in the closed forest than in open areas because of a larger surface area exposed to evaporation and a lower albedo when the snow is intercepted on branches in the forest (Leonard and Eschner, 1968; Brant, 1986).

Dunne and Leopold (1978) report that more than 30% to 50% of the gross precipitation can be intercepted in European coniferous forests leaving less for melt in the spring.

The snowmelt of 1986 comprised two runoff periods (Fig. 9(a)). The first melt which started at the end of March caused vertical movement of percolating water in the surface layers of the soil (humic horizon and the first mineral horizon). When a new frost period started in the middle of April, the percolated water refroze and created an ice front. On 21 April, the daily mean air temperature again exceeded 0°C and the final snowmelt started. At this time the temperature of the soil was still below 0°C and remained below 0°C until the whole catchment was snow-free on 28 April (cf. Fig. 11). The soil temperature then increased rapidly to a value close to the air temperature. Contribution of rain-on-snow and a large snow precipitation even just before the daily air temperature rose above 0°C released a large snowmelt runoff on frozen ground. Most of the runoff peak had passed before the soil temperature exceeded 0°C (Espeby, 1990a).

The very high runoff at the end of August 1986 (Fig. 9(a)) was initiated by a series of heavy summer rains starting a week before the largest on 30 August (53 mm) (Fig. 9(b)). This resulted in a flow of 7 mm day^{-1} if the whole catchment was considered to contribute to the flow measured at the V-notch. The accumulated measured runoff at the V-notch was 63.6 mm in 1986.

Snowpack simulation

The snowpack was not well simulated during the cold period of 1986, using the snow submodel in SOIL. A possible reason is that precipitation at the Sätra Gård station is collected from an open area and the snow model will, therefore, exaggerate snow production during the winter compared with that expected in a closed forest. The normal value used for the global radiation coefficient in the snowmelt function for forest had to be decreased (from a normal value for forests of 1.5×10^{-7} kg J^{-1} to a considerably lower value of 0.85×10^{-7} kg J^{-1}) in order to retard the melt and get a better fit with the water content in the forest snowpack measured at the end of the snowmelt

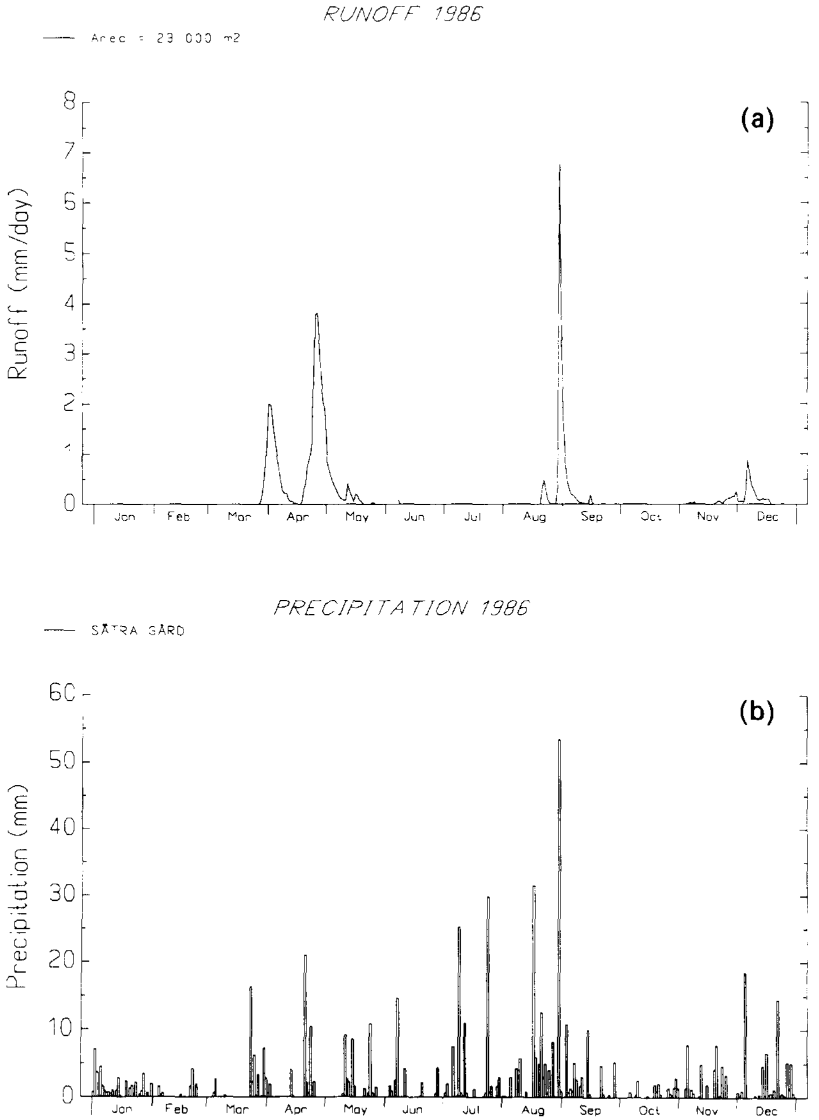


Fig. 9. (a) Measured runoff from Lund 1986. (b) Measured precipitation at Sätra Gård 1986.

(Fig. 10). This parameter may be as much as two to three times higher in open fields (Jansson, 1989), which implies that this forest is denser than normal. An increased interception storage capacity (i.e. denser forest) from 1.2 mm to 2.0–3.6 mm also helped to get a better fit.

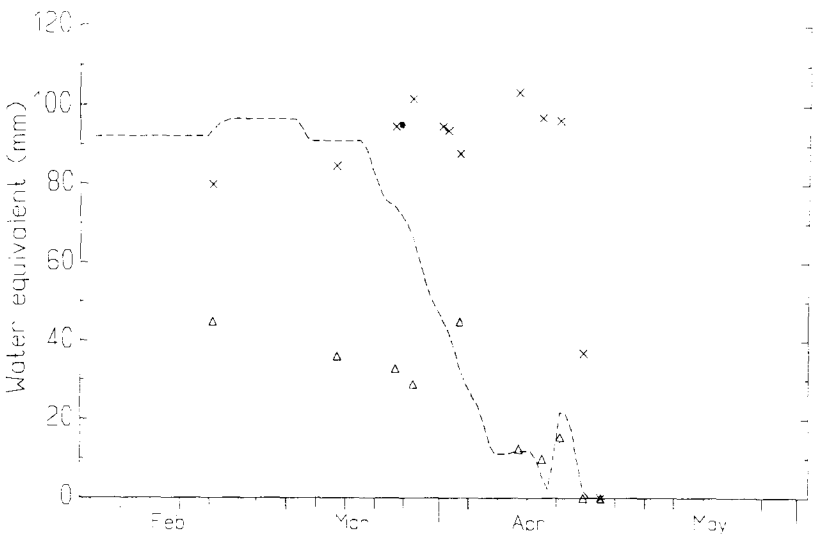


Fig. 10. Measured water equivalent of the snowpack in dense forest (△) and open sites (×) and simulated snowpack (dashed line) 1 February 1986–31 May 1986 with a global radiation coefficient of 0.85×10^{-7} kg J^{-1}.

Temperature simulation

For the snowmelt of 1986, temperature differences between measured (at soil pits Nos. 1 and 3) and simulated temperatures at three different depths, from two simulations of the slope realization, are shown in Fig. 11. One simulation is from the top level of the slope, with soil properties from pit No. 6 (simulation No. 1) and the second simulation is from the bottom level of the slope with soil properties from pit No. 1 (simulation No. 11). Just before the snowmelt the difference between simulated temperatures and measured temperatures is very small. The general trend observed is that the simulated temperatures rise faster than the measured soil temperatures and that the fit is very poor at and just after snowmelt. Generally the simulated temperatures are higher than measured, especially during the snowmelt. The difference between simulated and measured temperatures decreased with depth as an effect of damping. Both the simulated and measured temperatures at 7 cm show that the soil was frozen during the first part of the snowmelt.

This also coincides with field observations during the 1986 snowmelt (Espeby, 1990a). The simulated temperature increases more rapidly in the lower part of the catchment probably because of a higher water content as a result of a higher groundwater table.

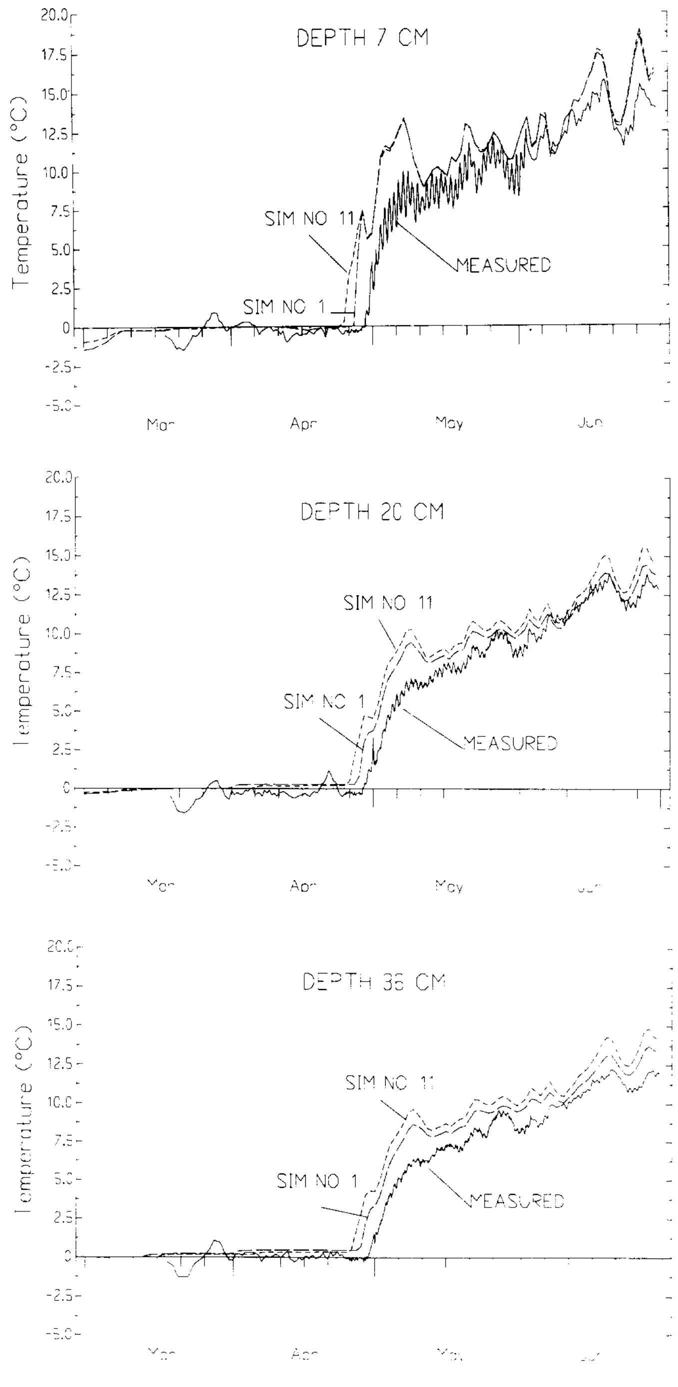


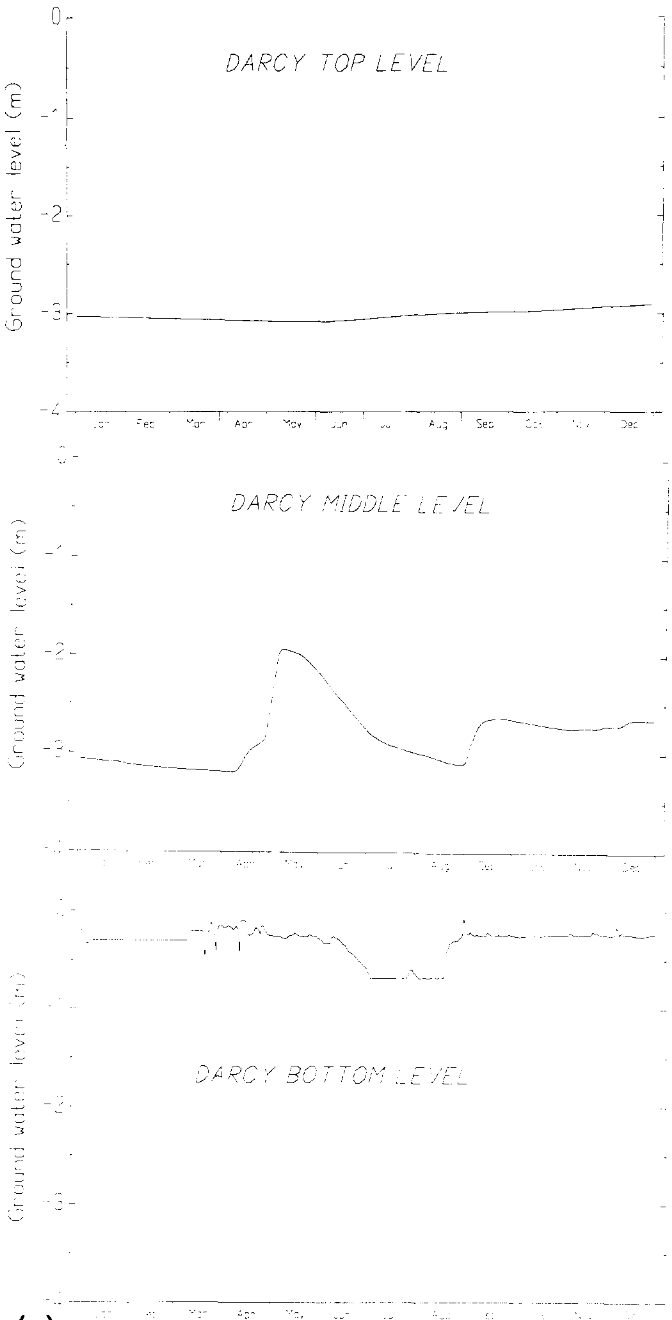
Fig. 11. A comparison between measured soil temperatures and simulated at three depths, 7, 20 and 36 cm from 1 March 1986 to 30 June 1986.

Slope simulation

The first slope realization (only matrix flow) indicated strongly that the runoff was governed by the soil physical properties found in the lower part of the catchment (soil pit No. 1). However, the measured runoff could not be fully simulated by assuming ordinary Darcian matrix flow, because it showed a more rapid response as well as recession and more peak flows even at small rain occasions (see Figs. 12 and 13). With only Darcian flow, a substantial amount of water is stored in the matrix and does not contribute to rapid runoff. As a consequence, the groundwater table remains low just before the autumn rains start at the end of August. Although smaller peaks in the measured runoff cannot be represented, most of the flow in SOIL is well described using the Darcy concept. The most important factor governing the characteristics of the runoff response is the hydraulic conductivity of the deepest layer at the bottom part of the slope, i.e. the dense basal till which has an average saturated hydraulic conductivity of 0.009 cm h^{-1}. This is based on a sensitivity analysis with varying conductivity of the deepest layer. The drainable porosity for this layer (determined as the difference between the total porosity and the field capacity value at a water tension of -10 kPa) was at the same time 8 vol.% (Espeby, 1990b).

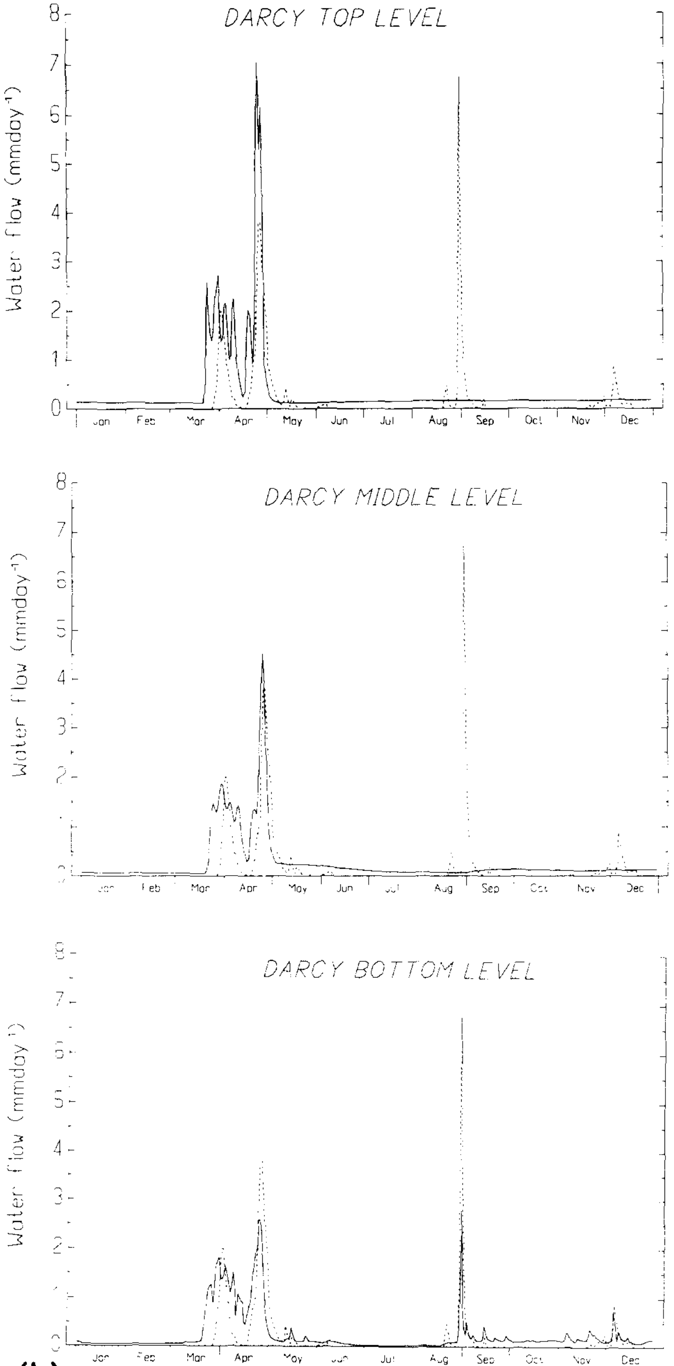
Three simulations from three levels in the slope (top, middle, and bottom levels) of a slope realization with Darcy and bypass flow accounted for in the model are shown in Figs. 12 and 13, respectively. At the bottom level of the slope the groundwater table is raised to the surface horizons where the horizons 0–10, 10–20, and 20–30 cm conduct the peak flow intensities. These three horizons include the highly macroporous humus-impregnated mineral horizon and the wave-washed horizon (cf. Fig. 3).

The groundwater table response is very slow in the upper part of the slope when a Darcian approach is used, but is changed dramatically with the bypass flow operating. The soil properties at the middle slope level reduced the flow peaks in response and the bypass scaling parameter A_{SC} had to be decreased with one order of magnitude (compared with two other levels in the slope) in order to get a quick response in the groundwater tables at this level. The A_{SC} in the bypass flow realization for simulation Nos. 1–5, 6–10, 11–15, were 0.135×10^{-1}, 0.250×10^{-2}, and 0.500×10^{-1}, respectively, in the simulations shown. Both realizations (pure Darcian and bypass flow) indicated that most of the water that was forming the runoff during snowmelt was generated from the upper slope levels. With the soil properties found at the upper level, snowmelt generated a surface runoff similar to that measured in the V-notch (Figs. 12, 13 and 14). With bypass flow, the surface runoff could be decreased letting more water infiltrate and elevate the groundwater



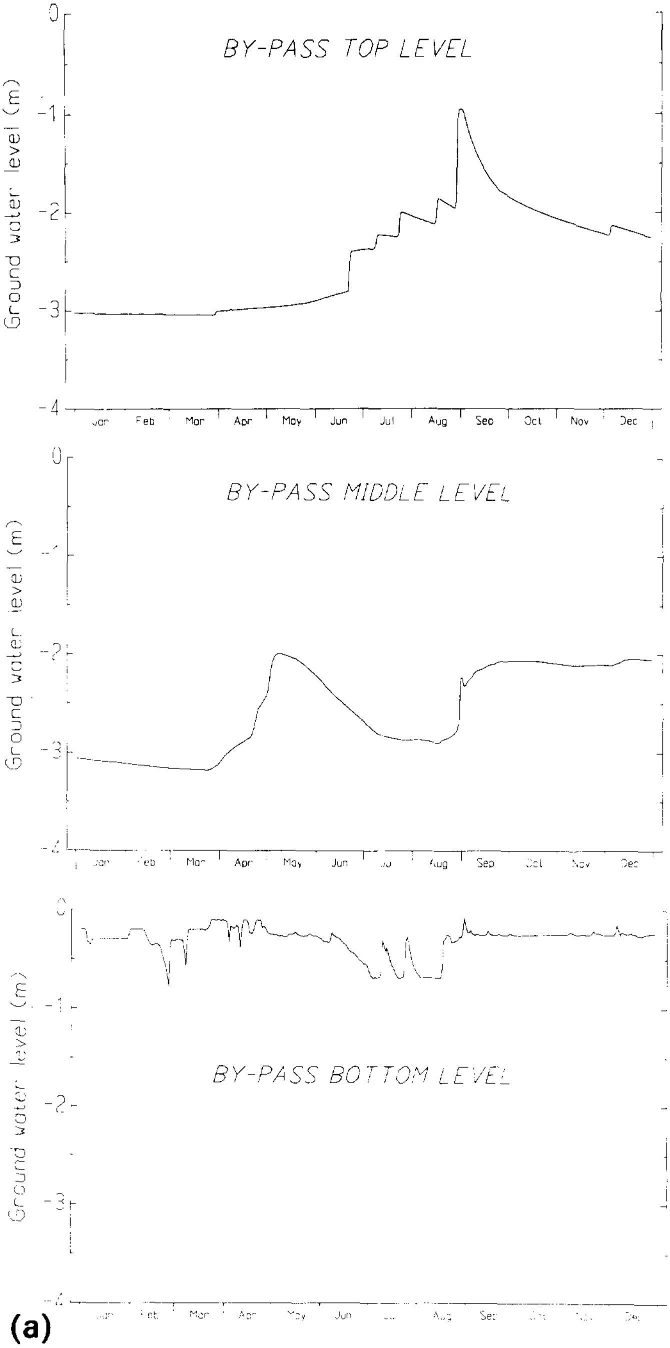
(a)

Fig. 12. (a) Simulated groundwater tables, (b) water flow (sum of model compartments with saturated output) 1 January 1986–31 December 1986, from three slope simulations (Nos. 1, 6 and 11) of a Darcian flow realization, compared with measured runoff (dashed line).



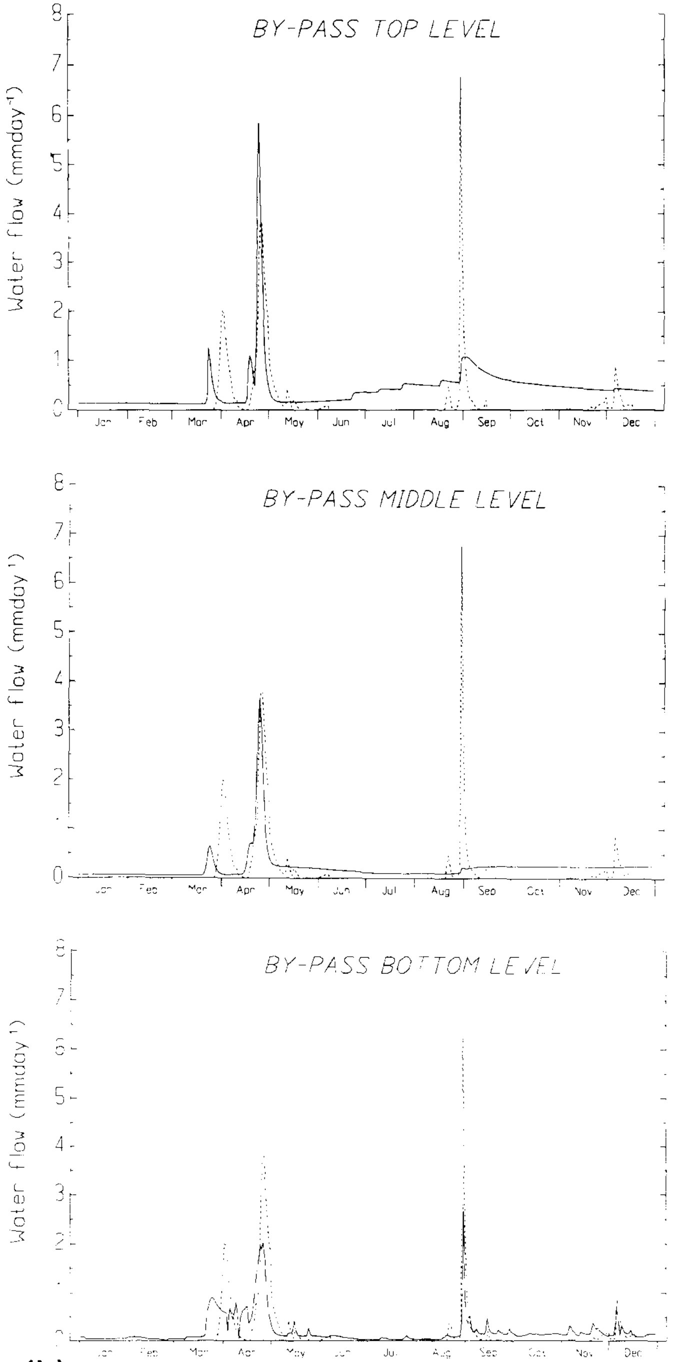
(b)

Fig. 12. Continued.



(a)

Fig. 13. (a) Simulated groundwater tables, and (b) water flow (sum of model compartments with saturated output) 1 January 1986–31 December 1986, from three simulations of (Nos. 1, 6 and 11) of a bypass flow realization, compared with measured runoff (dashed line).



(b)

Fig. 13. Continued.

TABLE 2

Accumulated water flow for different layers in the model for both Darcian and bypass flow after 11 simulations. Percentage of total simulated accumulated amount (Darcian, 89 mm and bypass, 73 mm) 1986

Layer	Darcy (%)	Bypass (%)
Surface runoff	35	16
0–10 cm	5	7
10–20 cm	29	36
20–30 cm	10	11
30–40 cm	8	10
Surface runoff–130 cm	92	87
0–40	52	64
130–450 cm	8	13

table (Figs. 12, 13 and 14). Field observations of melted snow spots indicate that a local seepage area and a spring situated in the ditch below soil pits 4 and 5 were the main contributing sources of runoff during the snowmelt (cf. Espeby, 1990a). The runoff during the autumn rains was only generated from the lower parts of the slope where the conductivity of the deepest layer was the most important factor regulating the discharge (cf. Figs. 12(b) and 13(b)). By accumulating the simulated flow in each layer from the bottom level of the slope (simulation No. 11) it was possible to see where the largest quantities were being conducted (see Table 2).

By adding the bypass flow in the model, much of the surface runoff produced in the Darcian simulation (35% of total water flow) during the snowmelt can be diverted to the porous layers including the macroporous humus-impregnated layer and the wave-washed gravelly sand layer beneath the humus horizons. Sixty-four percent of the flow occurs in the upper porous 40 cm. A consequence of the bypass flow is that more flow is available to produce a rapid response in rising groundwater tables. A comparison with the water balance is shown in Fig. 14, where the accumulated runoff from the slope has been adjusted for a period when the V-notch was not operative (before 20 March). The total accumulated amount of water flow from the model for 1986, was 89 mm and 73 mm for Darcy and bypass, respectively. The adjusted accumulated runoff for the same time period was 63 mm which is 75% of the Darcy flow and 87% of the bypass flow.

CONCLUSIONS

Simulation of water movement in till is difficult due to the problems involved in the determination of the water retention characteristic. Normally

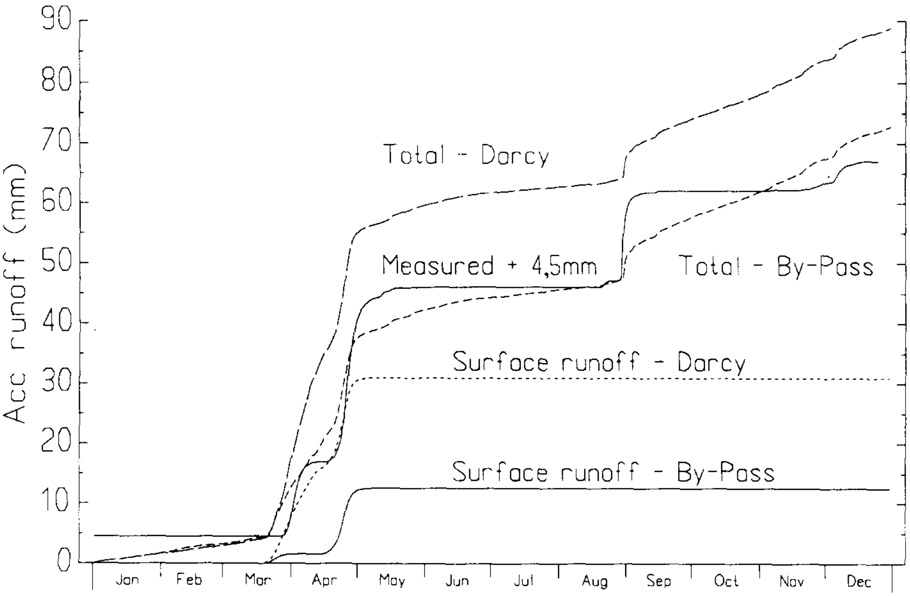


Fig. 14. Mean accumulated simulated surface runoff, mean accumulated runoff for 11 simulations compared with the accumulated runoff plus 4.5 mm (Darcian flow and bypass flow).

the pF-curve and the saturated hydraulic conductivity are determined on small sample volumes which cannot reflect the effect of boulders, cracks and other dominant structural features. These features mainly influence water movement during saturated conditions but also during occasions of unsaturated conditions just before intense snowmelt or heavy thundershowers.

The bypass scaling factor (A_{SC}) in this model is a very simple way of representing the macropore flow, although its physical basis can be questioned. It is not a directly measurable parameter nor does it account for the temporal variation in soil structure and macroporosity which may influence recharge (Jarvis and Jansson, 1989). A double porosity model would be more appropriate in terms of the physics of the problem, but the numerical requirements are more demanding for long-term transient simulations. This simulation study has also shown that it is possible to simulate a two-dimensional problem with a one-dimensional model.

It is necessary to choose a model with a physical basis, if the processes of flow are of interest. Although, the model used in this study was a physical model solving Richards' equation, the simulations have indicated that a Darcian approach cannot fully explain the water flow in a heterogeneous soil of this kind. The most important factor controlling the rapid response in the groundwater table and thus the saturated outflow was the saturated hydraulic conductivity and low water release properties of the dense unaffected basal till in the bottom part of the slope. This confirms the conclusions of another model study performed with a two-dimensional finite element model on a till slope under Swedish conditions (Johansson, 1985) where the soil parameters and infiltration at the slope base were influencing the quick response in groundwater outflow. The SOIL model was also very sensitive to changes in the hydraulic conductivity of the bottom layer in this slope base and an average value of the saturated hydraulic conductivity was the best approach to get a good representation with a Darcian concept. However, the simulated saturated outflow could not totally explain the measured runoff at the V-notch. Theoretical studies by Selim (1987) on the effect of anisotropy on outflow rates from a two-layer sloping area also confirm the fact that the discharging area decreases with increasing anisotropy of the top soil layer, resulting in a concentrated outflow at the bottom parts of the slope. With increasing anisotropy of the top soil layer the relative water flow rate will increase by at least one order of magnitude when the horizontal conductivity is four times higher than at isotropy (Selim, 1987). It is, in this case, important to state that the ratio of conductivity of the upper 60 cm of the soil to the lower basal till in the slope at Lund is up to 10 000.

The fact that field studies have shown that macropores and macrostructures were effective flow factors during occasions of snowmelt and heavy rains (Espeby, 1988, 1990a) gave support for incorporating bypass flow into the model. With the simple approach used in the SOIL model is was possible to get a quicker response in the groundwater tables even in smaller rain events and in this way to get a better agreement with the measured runoff from the slope.

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